JAMES MORRISON

Research Engineer

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GitHub: github.com/jmorrison-research

PROFESSIONAL SUMMARY

Results-driven and highly motivated Research Engineer with a proven track record in leveraging cutting-edge technologies to deliver actionable insights across mission-critical projects. Passionate about spearheading innovative solutions through data-driven decision making and cross-functional collaboration. Champion of synergistic approaches to drive organisational objectives whilst delivering value through dynamic problem-solving capabilities. Committed to continuous improvement and advancing organisational excellence through proven technical expertise.

WORK EXPERIENCE

Research Engineer

Quantum Labs Ltd

www.quantumlabs.io

Glasgow, United Kingdom

March 2022 - Present

- Designed and implemented machine learning algorithms for predictive analytics in distributed systems, achieving 34% improvement in model accuracy through rigorous experimentation and feature engineering

- Developed Python-based research frameworks and tooling to automate data processing pipelines, reducing manual intervention time by 22 hours per week across team workflows

- Collaborated with cross-functional teams of 8+ researchers and software engineers to define technical requirements and deliver two published research papers on neural network optimisation

- Maintained comprehensive documentation and conducted knowledge transfer sessions for 5 junior team members on advanced statistical methods and experimental design practices

Senior Software Developer

TechVision Solutions

www.techvisionsolutions.co.uk

Glasgow, United Kingdom

July 2020 - February 2022

- Engineered scalable microservices architecture using Docker and Kubernetes supporting 2 million daily active users, improving system latency by 41% and reducing infrastructure costs by 18%

- Led development of real-time data analytics platform processing 500GB+ daily data volumes using Apache Spark and Python, enabling stakeholders to extract actionable business insights

- Implemented comprehensive CI/CD pipelines utilising Jenkins and GitLab CI, reducing deployment time from 3 hours to 12 minutes and increasing team deployment frequency to 15+ releases per week

- Mentored 3 junior developers on software engineering best practices, code quality standards, and debugging methodologies resulting in 28% reduction in production incidents

Data Engineer

Insight Data Systems

www.insightdata.net

Edinburgh, United Kingdom

September 2018 - June 2020

- Built and optimised ETL pipelines processing complex datasets from 12+ heterogeneous sources using Python, SQL and Apache Airflow, ensuring 99.7% data accuracy and on-time delivery

- Developed data warehouse schema on PostgreSQL and Redis, supporting 150+ concurrent users with sub-second query response times for business intelligence dashboards

- Performed comprehensive database performance tuning and query optimisation, reducing average query execution time by 67% and improving overall system throughput

- Documented data lineage, created technical runbooks, and provided on-call support resulting in 95% issue resolution rate within agreed SLAs

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EDUCATION

BSc Computer Science (Hons)

University of Glasgow

Glasgow, United Kingdom

Graduated: June 2018

Grade: Upper Second Class Honours (2:1)

TECHNICAL SKILLS

Languages: Python, SQL, Java, JavaScript, Bash, R

Data Technologies: Apache Spark, Apache Airflow, PostgreSQL, Redis, Elasticsearch, Hadoop, Kafka

Machine Learning: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, XGBoost, Statistical Analysis

Cloud Platforms: AWS (EC2, S3, Lambda, RDS), Google Cloud Platform, Azure

DevOps and Tools: Docker, Kubernetes, Git, Jenkins, GitLab CI, Linux, Terraform

Web Frameworks: Flask, Django, REST APIs, FastAPI

Other: Jupyter Notebooks, Data Visualisation (Matplotlib, Plotly), Agile/Scrum, Experimental Design

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GITHUB REPOSITORIES

1. github.com/jmorrison-research/spark-ml-pipeline - Distributed machine learning pipeline framework for large-scale data processing

2. github.com/jmorrison-research/time-series-forecasting - Comprehensive toolkit for time-series analysis and ARIMA/Prophet model implementations

3. github.com/jmorrison-research/data-quality-framework - Automated data quality validation and anomaly detection system

4. github.com/jmorrison-research/neural-network-visualiser - Interactive PyTorch neural network architecture visualisation tool